

Telecom Laser Simulator Reference Manual

Equations, Parameter Tables, and Wave-Optical Formulations

1. Executive Summary

This reference manual specifies the mathematical and physical models implemented in the telecom diode laser simulator. It is designed to guide developers, semiconductor physicists, and device engineers. The model integrates 2D transverse carrier transport (drift-diffusion), unified wave optics (Helmholtz), and 1D longitudinal traveling wave feedback. In this release, the independent PhotonSolver is retired; photon populations are modeled directly through wave intensities and energy densities computed from EMSolver, ensuring absolute physical consistency.

2. Electrostatic & Drift-Diffusion Carrier Transport

Carrier transport in the 2D cross-section (y,z) is modeled via the Poisson equation coupled with the electron and hole continuity equations. The electrostatic potential (ψ) is determined by:

$$-\text{div}(\epsilon_0 \epsilon_r \text{grad}(\psi)) = q \cdot (p - n + N_D - N_A)$$

The electron and hole current densities, J_n and J_p , incorporate both drift (driven by the electric field $E_{\text{field}} = -\text{grad}(\psi)$) and diffusion (driven by concentration gradients):

$$J_n = -q \cdot \mu_n \cdot n \cdot \text{grad}(\psi) + q \cdot D_n \cdot \text{grad}(n)$$

$$J_p = -q \cdot \mu_p \cdot p \cdot \text{grad}(\psi) - q \cdot D_p \cdot \text{grad}(p)$$

Where:

- q is the elementary charge.
- μ_n and μ_p are the electron and hole mobilities.
- D_n and D_p are the diffusion coefficients, related to mobility by the Einstein relation under Boltzmann statistics: $D = \mu \cdot (k_B \cdot T / q)$.

The continuity equations govern the time-evolution and steady-state spatial profiles of carriers:

$$\text{div}(J_n) = q \cdot R_{\text{total}} \quad \text{and} \quad -\text{div}(J_p) = q \cdot R_{\text{total}}$$

where R_{total} is the sum of Shockley-Read-Hall (R_{SRH}), radiative spontaneous (R_{sp}), Auger (R_{Auger}), and stimulated (R_{stim}) recombination rates.

3. Unified Wave-Optical Solver & Photon Representation

To achieve full physical consistency, the simulator represents the photon population directly using the electromagnetic wave intensity of the guided mode computed by EMSolver, rather than tracking a separate scalar diffusion field.

The local photon density $s_{ph}(y,z)$ is derived by normalizing the transverse mode profile $E_{2d}(y,z)$ and scaling it with the average cavity power P_{cav} solved by the 1D longitudinal traveling wave equations:

$$s_{ph}(y,z) = [P_{cav} / (h * \nu * v_g)] * [|E_{2d}(y,z)|^2 / (\iint |E_{2d}(y,z)|^2 dy dz)]$$

Where:

- $h * \nu$ is the photon energy (transition energy E_{tr}).
- v_g is the optical group velocity in the waveguide core: $v_g = c / n_g$.
- $|E_{2d}(y,z)|^2$ is the transverse mode intensity.

The stimulated recombination rate R_{stim} is then directly formulated as:

$$R_{stim}(y,z) = v_g * g(y,z,\omega) * s_{ph}(y,z)$$

This ensures that stimulated carrier depletion (spatial hole burning) occurs only where the optical mode shape $|E_{2d}|^2$ overlaps with the gain profile g .

4. Complex Refractive Index & Helmholtz Solver

The transverse electromagnetic wave propagation is governed by the 2D complex Helmholtz equation:

$$[d^2/dy^2 + d^2/dz^2] E_{2d}(y,z) + k_0^2 * \epsilon_{tilde}(y,z,\omega) * E_{2d}(y,z) = \beta_{tilde}^2 * E_{2d}(y,z)$$

The complex relative permittivity ϵ_{tilde} is defined via the complex refractive index $n_{tilde} = n_r + i * n_i$ as: $\epsilon_{tilde} = n_{tilde}^2$. Using the approximation that $n_i \ll n_r$, we write:

$$\epsilon_{tilde}(y,z,\omega) = n_r(y,z,\omega)^2 + 2 * i * n_r(y,z,\omega) * n_i(y,z,\omega)$$

The real part $n_r(y,z,\omega)$ represents refraction and is modeled explicitly by material composition plus temperature (thermo-optic) and carrier (plasma dispersion) perturbations:

$$n_r(y,z,\omega) = n_{bulk}(y,z,\omega) + (dn_r/dT)*(T - T_0) + (dn_r/dn)*n$$

The imaginary part $n_i(y,z,\omega)$ represents localized gain and absorption losses. It is formulated via the explicit local gain $g(y,z,\omega)$, internal absorption $\alpha_i(y,z)$, and the uniform escape loss α_{escape} :

$$n_i(y,z,\omega) = -[c / (2 * \omega)] * [g(y,z,\omega) - \alpha_i(y,z) - \alpha_{escape}]$$

Substituting n_i back into the permittivity expression gives the final non-Hermitian operator profile:

$$\epsilon_{tilde}(y,z,\omega) = n_r^2 - i * [n_r * c / \omega] * [g(y,z,\omega) - \alpha_i(y,z) - \alpha_{escape}]$$

5. Material Parameter Tables (InP / InGaAsP System)

The following tables detail the typical parameter value ranges used for telecom laser simulations (1.31 μm / 1.55 μm) in the InP cladding and InGaAsP SCH / active MQW layers, along with literature references.

Parameter Name	InP Cladding	InGaAsP SCH/QW	Unit	Reference
Refractive Index (n_{bulk})	3.17 - 3.21	3.38 - 3.52	-	Piprek, Opto. Devices
Bandgap Energy (E_g)	1.35	0.75 - 0.95	eV	Vurgaftman et al. (2001)
Electron Mobility (μ_n)	3000 - 4500	2000 - 3500	$\text{cm}^2/\text{V-s}$	Coldren & Corzine
Hole Mobility (μ_p)	100 - 150	70 - 120	$\text{cm}^2/\text{V-s}$	Coldren & Corzine
SRH Recomb. Lifetime (τ)	2.0 - 5.0	1.0 - 3.0	ns	Piprek, Semiconductor Lasers
Radiative Coeff. (B_{sp})	1.0e-10	1.5e-10 - 2.5e-10	cm^3/s	Vurgaftman et al. (2001)
Auger Coefficient (C)	1.0e-30	1.0e-29 - 5.0e-29	cm^6/s	Coldren & Corzine
Thermo-Optic (dn/dT)	2.0e-4	2.5e-4	K-1	Piprek, Opto. Devices
Plasma Dispersion (dn/dn)	-1.8e-20	-2.5e-20	cm^3	Coldren & Corzine
Thermal Cond. (k_T)	0.68	0.05 - 0.12	W/cm-K	Piprek, Semiconductor Lasers

6. Numerical Coupling and Implementation Workflow

The solver iterates through a self-consistent loops scheduled as follows:

1. CarriersSolver: Computes ψ , n , and p based on drift-diffusion with Jacobian scaling.
2. Index and Gain Update: Evaluates the complex permittivity profile $\epsilon_{\text{tilde}}(y,z,\omega)$.
3. Helmholtz Solve (EMSolver): Solves the transverse modes and eigenvalues $\lambda = \beta_{\text{tilde}}^2$.
4. 1D Longitudinal Solve: Solves traveling wave equations to find cavity power P_{cav} and escape rate.
5. ThermoSolver: Solves lattice heat flow using Joule, recombination, and optical heating.
6. Convergence Check: The iteration terminates when field and carrier residuals fall below $1.0\text{e-}5$.

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